Math709 – Foundations of Computational Mathematics II

Spring 2013

Homework 2

1. Let $\mathbf{A}$ be an $m \times m$ matrix, let a_j be its j-th column. Prove the *Hadamard's inequality*:

$$|det\mathbf{A}| \leq \prod_{j=1}^{m} \|a_j\|_2.$$

2. Let $\mathbf{A}$ be a matrix with the property that columns $1, 3, 5, 7, \cdots$ are orthogonal to $2, 4, 6, 8, \cdots$. In a reduced QR factorization $\mathbf{A} = \hat{\mathbf{Q}}\hat{\mathbf{R}}$, what special structure does $\hat{\mathbf{R}}$ possess?

3. Determine the eigenvalues, determinant, and singular values of a Householder reflector. Additionally, for the eigenvalues, also give a geometric interpretation.

4. Consider the 2×2 orthogonal matrices

$$F = \begin{bmatrix} -c & s \\ s & c \end{bmatrix}, \quad J = \begin{bmatrix} c & s \\ -s & c \end{bmatrix},$$

where $s = \sin\theta$ and $c = \cos\theta$ for some θ. The first matrix F is a Householder reflector, and the second one J effects a rotation and is called *Givens rotation*.

   (a) Describe exactly what geometric effects left-multiplications by F and J have on the plane $\mathbb{R}^2$.

   (b) Describe an algorithm for QR factorization that is analogous to Algorithm 10.1 in Page 73, but based on Givens rotations instead of Household reflection.

   (c) Count the number of floating points involved the new algorithm proposed in (b) and compare it with that of Algorithm 10.1.

5. Suppose the $m \times n$ matrix $\mathbf{A}$ has the form

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{bmatrix},$$

where $\mathbf{A}_1$ is a nonsingular matrix of dimension $n \times n$ and $\mathbf{A}_2$ is an arbitrary matrix of dimension $(m-n) \times n$. Prove that $\|\mathbf{A}^+\|_2 \leq \|A_1^{-1}\|_2$.

6. (a) Write a MATLAB function **[Q,R] = mgs(A)** that computes a reduced QR factorization $\mathbf{A} = \hat{\mathbf{Q}}\hat{\mathbf{R}}$ of an $m \times n$ matrix $\mathbf{A}$ with $m \geq n$ using the Gram-Schmidt orthogonalization.

(b) Write a MATLAB function **[W,R] = house(A)** that computes an implicit representation of a full QR factorization $\mathbf{A} = \mathbf{QR}$ of an $m \times n$ matrix $\mathbf{A}$ with $m \geq n$ using Householder reflectors. The output variable $\mathbf{W}$ is a $m \times n$ matrix whose columns are the vectors v_k defining the successive Householder reflections. Then write a MATLAB function **Q = form(W)** that takes the matrix $\mathbf{W}$ produced by **house** as input and generates a corresponding orthogonal matrix $\mathbf{Q}$.

Print and test your programs by the reduced QR factorization of $\mathbf{Z}$ where

$$\mathbf{Z} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 7 \\ 4 & 2 & 3 \\ 4 & 2 & 2 \end{bmatrix}.$$

Also compare your results with the MATLAB's built-in command **[Q,R] = qr(Z,0)** and comment on any difference you see.

7. Let us consider the polynomial least squares fitting problem. Define t to be the m-vector corresponding to linearly spaced grid points from 0 to 1. Define $\mathbf{A}$ to be the $m \times n$ matrix associated with least squares fitting on t by a polynomial of degree $n-1$, and $\mathbf{b}$ to be the function $\cos(4t)$ evaluated on this grid. Set $m = 50$ and $n = 12$. Calculate and print (to sixteen-digit precision) the least squares coefficient vector $\mathbf{x}$ by the following methods:

(a) QR factorization computed by **mgs**.

(b) QR factorization computed by **house**.

(c) QR factorization computed by **qr**.

(d) Solution of the normal equation $\mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{A}^*\mathbf{b}$ using MATLAB's command \.

Comment on any differences you observe.